WorkFlow Global HRMS - Part 2 Report

PostgreSQL Database & Seed Data Implementation

1. Executive Summary

This report documents the implementation of Part 2 of the WorkFlow Global HRMS project, focusing on PostgreSQL database design, entity relationships, and seed data configuration. The implementation follows Entity Framework Core best practices and establishes a robust foundation for the HRMS application.

2. Database Architecture

The database architecture follows a modular approach with 12 feature-specific modules:

1. EmployeeFeature - Core employee management
2. AttendanceFeature - Time tracking and attendance
3. LeaveFeature - Leave requests and balances
4. PayrollFeature - Salary and payroll processing
5. DocumentFeature - Document management
6. ExpenseFeature - Expense reimbursements
7. PerformanceFeature - Goals and reviews
8. TrainingFeature - Training modules
9. RecruitmentFeature - Job postings and candidates
10. AnnouncementFeature - Company announcements
11. AnalyticsFeature - Metrics and analytics
12. CopilotFeature - AI assistant interactions

3. Entity Relationships & Normalization

All entities extend BaseEntity which provides:
- Id (Primary Key)
- CreatedOn, CreatedByUserId, CreatedByUserName
- ModifiedOn, ModifiedByUserId, ModifiedByUserName
- ProfilePicture
- DocumentType (for polymorphic queries)

Key Relationships:
- Employee -> AttendanceRecord (1:N)
- Employee -> LeaveRequest (1:N)
- Employee -> LeaveBalance (1:N)
- Employee -> PayrollRecord (1:N)
- Employee -> Document (1:N)
- Employee -> Reimbursement (1:N)
- Employee -> Goal (1:N)
- Employee -> PerformanceReview (1:N)
- JobPosting -> Candidate (1:N)

4. Database Indexes

Strategic indexes have been created for optimal query performance:

Employee Table:
- Email (Unique)
- ManagerId, Department, Status, Role

AttendanceRecord Table:
- EmployeeId, Date, Status
- Composite: (EmployeeId, Date) Unique

LeaveRequest Table:
- EmployeeId, Status, StartDate

LeaveBalance Table:
- EmployeeId
- Composite: (EmployeeId, LeaveType) Unique

PayrollRecord Table:
- EmployeeId, PayPeriodStart, Status

Document Table:
- EmployeeId, Category, Status

Reimbursement Table:
- EmployeeId, Status, Date

Goal & PerformanceReview Tables:
- EmployeeId, ReviewerId, Status

5. Seed Data Configuration

The system includes comprehensive seed data for demonstration:

Employees (8 users):
- 1 Admin (admin@workflowglobal.com)
- 1 HR Specialist (hr@workflowglobal.com)
- 1 Manager (manager@workflowglobal.com)
- 5 Employees across different departments

Attendance Records:
- 7 days of attendance history for sample employee
- Includes clock-in/out times, productive hours

Leave Balances:
- Default balances for all employees
- Casual (12), Sick (10), Personal (5) days

Training Modules:
- Security Awareness (Mandatory)
- Leadership Fundamentals
- Advanced C# Programming

Announcements:
- Welcome message
- Holiday notices

6. Demo Login Credentials

Email	Password	Role	Access Level
admin@workflowglobal.com	admin123	Admin	Full System
hr@workflowglobal.com	hr123	HR	HR Functions
manager@workflowglobal.com	manager123	Manager	Team Management
john.doe@workflowglobal.com	employee123	Employee	Self Service

7. Data Normalization & Best Practices

The database design follows normalization principles:

1. First Normal Form (1NF):
   - All tables have atomic values
   - No repeating groups

2. Second Normal Form (2NF):
   - All non-key attributes depend on primary key
   - Separate tables for different entities

3. Third Normal Form (3NF):
   - No transitive dependencies
   - LeaveBalance separated from LeaveRequest
   - Employee details not duplicated

Additional Best Practices:
- Consistent naming conventions
- Appropriate data types and precision
- Foreign key relationships
- Audit fields on all entities
- Enum types for status fields
- Soft delete capability via Status field

8. Migration Strategy

Entity Framework Core Migrations:

To apply migrations:
1. Navigate to API project directory
2. Run: dotnet ef migrations add InitialCreate
3. Run: dotnet ef database update

The migrations will:
- Create all 15+ database tables
- Apply indexes and constraints
- Seed initial data
- Configure relationships

Note: Ensure PostgreSQL connection string is configured in appsettings.json

9. Conclusion

Part 2 implementation successfully establishes:

- Comprehensive database schema for 12 HRMS modules
- Proper entity relationships and normalization
- Strategic indexing for performance
- Rich seed data for demonstration
- Entity Framework Core configurations
- Migration-ready database structure

The database foundation is now ready for:

- Authentication and authorization (Part 3)
- Business logic implementation
- GraphQL API development
- Frontend integration

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